PANDUAN INTEGRASI DATA

Web App → Microsoft Excel

ERP-ACC (Supabase/PostgreSQL) & BUL-Accounting (Firebase/Firestore)

Query Reference & Export Guide

Dibuat: 28 Mei 2026 · Versi 1.0

BUL-Accounting

Backend: Firebase Firestore
Framework: React 18 + Tailwind
Export: xlsx (sudah tersedia)
Koleksi: 9 koleksi aktif

ERP-ACC

Backend: Supabase (PostgreSQL)
Framework: React 18 + Ant Design
Export: belum ada, perlu ditambahkan
Tabel: 25+ tabel relasional

DAFTAR ISI

1. Analisa Kemungkinan Integrasi Excel

Dokumen ini menganalisa dua web app yang berjalan dalam satu repositori: **BUL-Accounting** (Firebase/Firestore) dan **ERP-ACC** (Supabase/PostgreSQL). Keduanya dapat dihubungkan ke Microsoft Excel melalui pendekatan yang berbeda sesuai dengan backend masing-masing.

1.1 BUL-Accounting — Status Export Saat Ini

Status: Export Excel SUDAH ADA. Library **xlsx** sudah dipasang dan digunakan di *src/utils/exportUtils.js*. Tersedia: export jurnal, neraca, dan laporan laba rugi ke format .xlsx.

Yang sudah tersedia di exportUtils.js:

Fungsi	Output File	Konten
exportJournalsToExcel()	Jurnal_Umum.xlsx	Tanggal, No.Jurnal, Keterangan, Debit, Kredit
exportNeracaToExcel()	Neraca_[tanggal].xlsx	Neraca posisi keuangan (Aset, Kewajiban, Ekuitas)
exportLabaRugiToExcel()	LabaRugi_[periode].xlsx	Laba Rugi lengkap dengan subtotal
exportToExcel()	[custom].xlsx	Generic — bisa dipakai untuk data apapun
exportToPDF()	[custom].pdf	Export tabel generik ke PDF

Yang belum tersedia dan perlu ditambahkan:

- Export data Penjualan (invoices) ke Excel
- Export data Kas/Bank (kas_bank) ke Excel
- Export data Biaya (biaya) ke Excel
- Export data Armada (armada) ke Excel
- Export data Aset (assets) ke Excel
- Export data Pelanggan & Supplier ke Excel

1.2 ERP-ACC — Status Export Saat Ini

Status: Export Excel BELUM ADA. ERP-ACC menggunakan Supabase (PostgreSQL) dengan 25+ tabel relasional. Perlu ditambahkan library xlsx dan fungsi export untuk setiap modul.

Modul ERP-ACC yang memerlukan export Excel:

Modul	Tabel Utama	Prioritas
Sales Orders & GD	sales_orders, goods_deliveries	Tinggi
Sales Invoices	invoices (type=sales)	Tinggi
Purchase Orders & GR	purchase_orders, goods_receipts	Tinggi
Purchase Invoices	invoices (type=purchase)	Tinggi
Pembayaran (Cash/Bank)	payments, accounts	Tinggi
Jurnal Akuntansi	journals, journal_items, coa	Sedang
Laporan (Neraca, L/R)	coa, journal_items	Sedang
Stok & Kartu Stok	inventory_stock, inventory_movements	Sedang

Modul	Tabel Utama	Prioritas
Aset Tetap	assets, asset_categories	Rendah
Sales & Purchase Returns	sales_returns, purchase_returns	Rendah

1.3 Arsitektur Integrasi yang Direkomendasikan

Pendekatan	Cocok Untuk	Pro	Kontra
In-App Button Export (xlsx di browser)	Kedua app	Tidak perlu server, langsung dari UI	Terbatas filter, data real-time
Supabase REST API + Power Query Excel	ERP-ACC	Excel bisa refresh otomatis, no-code	Perlu koneksi internet, perlu API key
Node.js Script (export semua data)	Kedua app	Bisa dijadwalkan, export lengkap	Perlu akses server, setup sekali
Google Sheets (via Firestore SDK)	BUL-Accounting	Real-time sync, bisa dibagikan	Perlu Apps Script, setup lebih kompleks

Rekomendasi: Untuk solusi paling cepat, tambahkan tombol "Export Excel" di setiap halaman daftar menggunakan library **xlsx** (sudah dipasang di bul-accounting, perlu install di erp-acc). Untuk ERP-ACC, Supabase Power Query adalah opsi terbaik untuk koneksi langsung ke Excel.

2. BUL-Accounting — Koleksi Firestore

2.1 Daftar Koleksi & Struktur Field

Koleksi: journals

Jurnal umum double-entry. Ini adalah sumber utama semua data akuntansi.

Field	Tipe	Keterangan
id	string	Auto-generated Firestore document ID
date	string (ISO)	Tanggal jurnal, mis. "2026-01-15"
description	string	Keterangan/deskripsi jurnal
type	string	Jenis: penjualan, biaya, kas, dll.
truckId	string|null	ID armada/truck terkait
status	string	"posted" | "deleted"
totalDebit	number	Total debit (harus = totalCredit)
totalCredit	number	Total kredit
lines	array	[{accountCode, debit, credit, description}]
createdAt	string (ISO)	Timestamp dibuat
createdBy	string	UID user yang membuat

Koleksi: invoices

Faktur penjualan ke pelanggan.

Field	Tipe	Keterangan
id	string	Auto-generated Firestore document ID
date	string (ISO)	Tanggal faktur
invoiceNo	string	Nomor faktur
customerId	string	ID pelanggan
customerName	string	Nama pelanggan (denormalized)
truckId	string|null	ID armada terkait
amount	number	Jumlah tagihan
status	string	"draft"|"unpaid"|"partial"|"paid"|"cancelled"
description	string	Keterangan

Koleksi: customers (pelanggan)

Master data pelanggan.

Field	Tipe	Keterangan
id	string	Auto-generated Firestore document ID
name	string	Nama pelanggan
address	string	Alamat
phone	string	Nomor telepon
email	string	Email
isActive	boolean	Status aktif (soft-delete)

Koleksi: kas_bank

Transaksi kas dan bank.

Field	Tipe	Keterangan
id	string	Auto-generated Firestore document ID
date	string (ISO)	Tanggal transaksi
type	string	"masuk" | "keluar"
amount	number	Jumlah
accountCode	string	Kode akun kas/bank
description	string	Keterangan
category	string	Kategori transaksi

Koleksi: biaya

Catatan beban/biaya operasional.

Field	Tipe	Keterangan
id	string	Auto-generated Firestore document ID
date	string (ISO)	Tanggal biaya
description	string	Keterangan biaya
amount	number	Jumlah biaya
accountCode	string	Kode akun biaya
truckId	string|null	ID armada (opsional)

Koleksi: armada

Master data armada/truck.

Field	Tipe	Keterangan
id	string	Auto-generated Firestore document ID
plateNumber	string	Nomor polisi
type	string	Jenis armada
isActive	boolean	Status aktif

Koleksi: assets

Aset tetap perusahaan.

Field	Tipe	Keterangan
id	string	Auto-generated Firestore document ID
name	string	Nama aset
acquisitionDate	string	Tanggal perolehan
acquisitionCost	number	Harga perolehan
usefulLife	number	Masa manfaat (bulan)
accumulatedDepreciation	number	Akumulasi penyusutan

Koleksi: audit_log

Log audit setiap perubahan jurnal.

Field	Tipe	Keterangan
journalId	string	ID jurnal terkait
action	string	"create"|"update"|"delete"
by	string	User yang melakukan aksi
at	string (ISO)	Timestamp aksi

2.2 Firestore JavaScript Queries

Query-query berikut dapat digunakan langsung di JavaScript/Node.js untuk mengambil data dari Firestore. Import yang diperlukan:

```
import { db } from '../firebase'
import { collection, query, where, orderBy, getDocs } from 'firebase/firestore'
```

Q-BUL-01: Semua Jurnal yang Diposting

Ambil semua jurnal dengan status posted, diurutkan by tanggal.

```
// Q-BUL-01: Semua Jurnal Posted
async function getAllJournals() {
const q = query(
collection(db, 'journals'),
where('status', '==', 'posted'),
orderBy('date', 'desc')
)
const snap = await getDocs(q)
return snap.docs.map(d => ({ id: d.id, ...d.data() }))
}
```

Q-BUL-02: Jurnal per Periode

Filter jurnal berdasarkan rentang tanggal (client-side filtering untuk Firestore free tier).

```
// Q-BUL-02: Jurnal per Periode
async function getJournalsByPeriod(startDate, endDate) {
const snap = await getDocs(query(
collection(db, 'journals'),
where('status', '==', 'posted')
))
return snap.docs
.map(d => ({ id: d.id, ...d.data() }))
.filter(j => j.date >= startDate && j.date <= endDate)
.sort((a, b) => a.date.localeCompare(b.date))
}
```

Q-BUL-03: Jurnal per Truck/Armada

Filter jurnal untuk armada tertentu (useful untuk laporan per truck).

```
// Q-BUL-03: Jurnal per Armada
async function getJournalsByTruck(truckId) {
const q = query(
collection(db, 'journals'),
where('status', '==', 'posted'),
where('truckId', '==', truckId)
)
const snap = await getDocs(q)
return snap.docs.map(d => ({ id: d.id, ...d.data() }))
}
```

Q-BUL-04: Semua Invoice/Penjualan

Ambil semua faktur penjualan aktif.

```
// Q-BUL-04: Semua Invoice Penjualan
async function getAllInvoices() {
const q = query(
collection(db, 'invoices'),
orderBy('date', 'desc')
)
const snap = await getDocs(q)
return snap.docs.map(d => ({ id: d.id, ...d.data() }))
}
```

Q-BUL-05: Invoice Belum Lunas (Outstanding AR)

Filter invoice yang masih outstanding (unpaid atau partial).

```
// Q-BUL-05: Invoice Belum Lunas
async function getOutstandingInvoices() {
const snap = await getDocs(query(
collection(db, 'invoices')
))
return snap.docs
.map(d => ({ id: d.id, ...d.data() }))
.filter(inv => ['unpaid', 'partial'].includes(inv.status))
.sort((a, b) => a.date.localeCompare(b.date))
}
```

Q-BUL-06: Data Rekap Akun (Saldo per Akun)

Hitung saldo setiap akun dari semua jurnal — digunakan untuk neraca & laba rugi.

```
// Q-BUL-06: Rekap Saldo per Akun
async function getAccountBalances(endDate) {
const journals = await getJournalsByPeriod('1900-01-01', endDate)
const balances = {}
journals.forEach(j => {
j.lines?.forEach(line => {
if (!balances[line.accountCode])
balances[line.accountCode] = { debit: 0, credit: 0 }
balances[line.accountCode].debit += (line.debit || 0)
balances[line.accountCode].credit += (line.credit || 0)
})
})
return balances // { '1111': { debit: X, credit: Y }, ... }
}
```

Q-BUL-07: Semua Data Kas/Bank

Ambil transaksi kas dan bank.

```
// Q-BUL-07: Transaksi Kas/Bank
async function getKasBank(startDate, endDate) {
const snap = await getDocs(query(
collection(db, 'kas_bank'),
orderBy('date', 'desc')
))
let results = snap.docs.map(d => ({ id: d.id, ...d.data() }))
if (startDate) results = results.filter(r => r.date >= startDate)
if (endDate) results = results.filter(r => r.date <= endDate)
return results
}
```

Q-BUL-08: Semua Data Biaya

Ambil semua biaya operasional.

```
// Q-BUL-08: Data Biaya
async function getAllBiaya(startDate, endDate) {
const snap = await getDocs(collection(db, 'biaya'))
let results = snap.docs.map(d => ({ id: d.id, ...d.data() }))
if (startDate) results = results.filter(r => r.date >= startDate)
if (endDate) results = results.filter(r => r.date <= endDate)
return results.sort((a,b) => a.date.localeCompare(b.date))
}
```

2.3 Kode Export Excel — BUL-Accounting (Siap Pakai)

Fungsi-fungsi di bawah dapat langsung ditambahkan ke *src/utils/exportUtils.js*. Library **xlsx** sudah tersedia. Cukup import dan panggil dari komponen yang sesuai.

Export Penjualan (Invoice) ke Excel

```
// Tambahkan ke exportUtils.js
export function exportPenjualanToExcel(invoices, filename = 'Penjualan') {
const rows = invoices.map(inv => ({
'Tanggal': inv.date,
'No. Invoice': inv.invoiceNo || '-',
'Pelanggan': inv.customerName || inv.customerId,
'Armada': inv.truckId || '-',
'Jumlah': inv.amount || 0,
'Status': inv.status,
'Keterangan': inv.description || '',
}))
const ws = XLSX.utils.json_to_sheet(rows)
ws['!cols'] = [
{wch:12},{wch:16},{wch:25},{wch:15},{wch:18},{wch:12},{wch:30}
]
const wb = XLSX.utils.book_new()
XLSX.utils.book_append_sheet(wb, ws, 'Penjualan')
XLSX.writeFile(wb, `${filename}.xlsx`)
}
```

Export Jurnal Lengkap dengan Detail Lines

```
export function exportJurnalDetailToExcel(journals, filename = 'Jurnal_Detail') {
const rows = []
journals.forEach((j, idx) => {
j.lines?.forEach((line, li) => {
rows.push({
'No': li === 0 ? idx + 1 : '',
'Tanggal': li === 0 ? j.date : '',
'No.Jurnal': li === 0 ? j.id.slice(0,8).toUpperCase() : '',
'Keterangan': li === 0 ? j.description : '',
'Tipe': li === 0 ? (j.type || '-') : '',
'Armada': li === 0 ? (j.truckId || '-') : '',
'Kode Akun': line.accountCode,
'Debit': line.debit || 0,
'Kredit': line.credit || 0,
'Ket. Baris': line.description || '',
})
})
rows.push({}) // separator
})
const ws = XLSX.utils.json_to_sheet(rows)
ws['!cols'] = [
{wch:4},{wch:12},{wch:12},{wch:35},{wch:14},
{wch:14},{wch:12},{wch:18},{wch:18},{wch:25}
]
const wb = XLSX.utils.book_new()
XLSX.utils.book_append_sheet(wb, ws, 'Jurnal')
XLSX.writeFile(wb, `${filename}.xlsx`)
}
```

Export Semua Koleksi Sekaligus (Multi-Sheet)

```
export async function exportAllDataToExcel(db, filename = 'BUL_All_Data') {
const wb = XLSX.utils.book_new()
// Sheet 1: Jurnal
const journals = await getAllJournals(db)
const jRows = []
journals.forEach(j => j.lines?.forEach(line => jRows.push({
Tanggal: j.date, Jurnal: j.id.slice(0,8),
Keterangan: j.description, Akun: line.accountCode,
Debit: line.debit || 0, Kredit: line.credit || 0
})))
XLSX.utils.book_append_sheet(wb, XLSX.utils.json_to_sheet(jRows), 'Jurnal')
// Sheet 2: Penjualan
const invoices = await getAllInvoices(db)
const iRows = invoices.map(inv => ({
Tanggal: inv.date, Invoice: inv.invoiceNo,
Pelanggan: inv.customerName, Jumlah: inv.amount, Status: inv.status
}))
XLSX.utils.book_append_sheet(wb, XLSX.utils.json_to_sheet(iRows), 'Penjualan')
XLSX.writeFile(wb, `${filename}.xlsx`)
}
```

3. ERP-ACC — Tabel Supabase (PostgreSQL)

3.1 Peta Relasi Tabel Utama

Tabel	Relasi Utama	Deskripsi
customers	→ invoices, sales_orders, payments	Master data pelanggan
suppliers	→ invoices, purchase_orders, payments	Master data supplier
products	→ sales/purchase_order_items, inventory_stock	Master produk/barang
units	→ products, order_items	Satuan ukuran (kg, ton, m3, dll.)
coa	→ journal_items	Chart of Accounts (akun buku besar)
tax_codes	→ products, invoices	Kode pajak (PPN, PPh, dll.)
payment_terms	→ sales_orders, purchase_orders	Termin pembayaran
warehouses	→ orders, goods_docs, stock	Gudang/lokasi stok
sales_orders	→ goods_deliveries, invoices	Order penjualan
sales_order_items	← sales_orders	Line item SO
goods_deliveries	→ invoices (linked)	Surat jalan pengiriman
goods_delivery_items	← goods_deliveries	Line item GD
purchase_orders	→ goods_receipts, invoices	Order pembelian
purchase_order_items	← purchase_orders	Line item PO
goods_receipts	→ invoices (linked)	Penerimaan barang
goods_receipt_items	← goods_receipts	Line item GR
invoices	→ payments, journals	Faktur (sales & purchase)
invoice_items	← invoices	Line item faktur
payments	→ journals, accounts	Pembayaran AR/AP
journals	→ journal_items	Header jurnal akuntansi
journal_items	← journals, coa	Baris debit/kredit jurnal
accounts	→ payments, transfers	Rekening kas & bank
assets	→ asset_categories, journals	Aset tetap
asset_categories	→ assets	Kategori aset tetap
inventory_stock	← products, warehouses	Saldo stok per produk
inventory_movements	← products, orders	Mutasi stok
sales_returns	→ sales_orders	Return penjualan
sales_return_items	← sales_returns	Line item return penjualan
purchase_returns	→ purchase_orders	Return pembelian
purchase_return_items	← purchase_returns	Line item return pembelian

3.2 SQL Queries untuk Export Excel

Query berikut dijalankan langsung di Supabase SQL Editor (Database → SQL Editor) atau melalui PostgreSQL client. Ganti *[YYYY-MM-DD]* dengan tanggal yang diinginkan.

Q-ERP-01 — Export Sales Orders Lengkap

Semua Sales Order dengan nama pelanggan, status, nilai, dan payment term.

```
-- Q-ERP-01: Export Sales Orders
SELECT
so.id,
so.so_number AS "No SO",
so.date AS "Tanggal",
c.name AS "Pelanggan",
pt.name AS "Termin Bayar",
w.name AS "Gudang",
so.subtotal,
so.tax_amount AS "PPN",
so.total AS "Total",
so.status,
so.notes AS "Catatan",
so.created_at
FROM sales_orders so
LEFT JOIN customers c ON c.id = so.customer_id
LEFT JOIN payment_terms pt ON pt.id = so.payment_term_id
LEFT JOIN warehouses w ON w.id = so.warehouse_id
ORDER BY so.date DESC, so.created_at DESC;
```

Q-ERP-02 — Export Sales Orders + Item Lines (Detail)

Detail baris per produk dari setiap Sales Order.

```
-- Q-ERP-02: Sales Orders dengan Item Detail
SELECT
so.so_number AS "No SO",
so.date AS "Tanggal SO",
c.name AS "Pelanggan",
so.status AS "Status SO",
p.sku AS "SKU",
p.name AS "Produk",
u.name AS "Satuan",
soi.quantity AS "Qty",
soi.unit_price AS "Harga",
soi.tax_amount AS "Pajak",
soi.total AS "Subtotal Baris"
FROM sales_orders so
JOIN sales_order_items soi ON soi.sales_order_id = so.id
JOIN customers c ON c.id = so.customer_id
JOIN products p ON p.id = soi.product_id
JOIN units u ON u.id = soi.unit_id
ORDER BY so.date DESC, so.so_number, p.name;
```

Q-ERP-03 — Export Sales Invoices (Faktur Penjualan)

Semua faktur penjualan beserta status pembayaran dan sisa tagihan.

```
-- Q-ERP-03: Sales Invoices
SELECT
inv.invoice_number AS "No Faktur",
inv.date AS "Tanggal",
inv.due_date AS "Jatuh Tempo",
c.name AS "Pelanggan",
so.so_number AS "Ref SO",
inv.subtotal,
inv.tax_amount AS "PPN",
inv.total AS "Total",
inv.amount_paid AS "Sudah Dibayar",
(inv.total - inv.amount_paid) AS "Sisa Tagihan",
inv.status,
CASE
WHEN inv.due_date < CURRENT_DATE
AND inv.status NOT IN ('paid','cancelled')
THEN 'JATUH TEMPO'
ELSE '-'
END AS "Overdue"
FROM invoices inv
JOIN customers c ON c.id = inv.customer_id
LEFT JOIN sales_orders so ON so.id = inv.sales_order_id
WHERE inv.type = 'sales'
ORDER BY inv.date DESC;
```

Q-ERP-04 — Export Purchase Orders

Semua Purchase Order dengan supplier dan nilai.

```
-- Q-ERP-04: Purchase Orders
SELECT
po.po_number AS "No PO",
po.date AS "Tanggal",
s.name AS "Supplier",
pt.name AS "Termin Bayar",
w.name AS "Gudang",
po.subtotal,
po.tax_amount AS "PPN",
po.total AS "Total",
po.status,
po.notes AS "Catatan"
FROM purchase_orders po
LEFT JOIN suppliers s ON s.id = po.supplier_id
LEFT JOIN payment_terms pt ON pt.id = po.payment_term_id
LEFT JOIN warehouses w ON w.id = po.warehouse_id
ORDER BY po.date DESC;
```

Q-ERP-05 — Export Purchase Invoices (Hutang Usaha)

Faktur pembelian dengan status dan sisa hutang ke supplier.

```
-- Q-ERP-05: Purchase Invoices / Hutang Usaha
SELECT
inv.invoice_number AS "No Faktur",
inv.date AS "Tanggal",
inv.due_date AS "Jatuh Tempo",
s.name AS "Supplier",
po.po_number AS "Ref PO",
inv.total AS "Total",
inv.amount_paid AS "Sudah Dibayar",
(inv.total - inv.amount_paid) AS "Sisa Hutang",
inv.status,
CASE
WHEN inv.due_date < CURRENT_DATE
AND inv.status NOT IN ('paid','cancelled')
THEN 'JATUH TEMPO'
ELSE '-'
END AS "Overdue"
FROM invoices inv
JOIN suppliers s ON s.id = inv.supplier_id
LEFT JOIN purchase_orders po ON po.id = inv.purchase_order_id
WHERE inv.type = 'purchase'
ORDER BY inv.date DESC;
```

Q-ERP-06 — Export Pembayaran (Penerimaan & Pengeluaran)

Semua transaksi pembayaran melalui kas/bank.

```
-- Q-ERP-06: Semua Pembayaran
SELECT
p.date AS "Tanggal",
p.type AS "Tipe",
COALESCE(c.name, s.name) AS "Pihak",
acc.name AS "Rekening",
inv.invoice_number AS "Ref Faktur",
p.amount AS "Jumlah",
p.notes AS "Catatan",
p.created_at AS "Dibuat Pada"
FROM payments p
LEFT JOIN customers c ON c.id = p.customer_id
LEFT JOIN suppliers s ON s.id = p.supplier_id
LEFT JOIN accounts acc ON acc.id = p.account_id
LEFT JOIN invoices inv ON inv.id = p.invoice_id
ORDER BY p.date DESC;
```

Q-ERP-07 — Export Jurnal Akuntansi Lengkap

Semua jurnal dengan baris debit/kredit dan kode akun COA.

```
-- Q-ERP-07: Jurnal Akuntansi Detail
SELECT
j.journal_number AS "No Jurnal",
j.date AS "Tanggal",
j.source AS "Sumber",
j.description AS "Keterangan",
j.is_posted AS "Diposting",
coa.code AS "Kode Akun",
coa.name AS "Nama Akun",
ji.debit,
ji.credit,
ji.description AS "Ket. Baris"
FROM journals j
JOIN journal_items ji ON ji.journal_id = j.id
JOIN coa ON coa.id = ji.coa_id
WHERE j.is_posted = true
ORDER BY j.date DESC, j.journal_number, coa.code;
```

Q-ERP-08 — Laporan Buku Besar per Akun

Riwayat mutasi satu akun dengan saldo berjalan (running balance).

```
-- Q-ERP-08: Buku Besar Akun (ganti '1111' dengan kode akun yang diinginkan)
WITH ledger AS (
SELECT
j.date,
j.journal_number,
j.description,
ji.debit,
ji.credit
FROM journal_items ji
JOIN journals j ON j.id = ji.journal_id
JOIN coa ON coa.id = ji.coa_id
WHERE coa.code = '1111' -- <-- ganti kode akun
AND j.is_posted = true
ORDER BY j.date, j.journal_number
)
SELECT
date AS "Tanggal",
journal_number AS "No Jurnal",
description AS "Keterangan",
debit AS "Debit",
credit AS "Kredit",
SUM(debit - credit) OVER (ORDER BY date, journal_number)
AS "Saldo"
FROM ledger;
```

Q-ERP-09 — Laporan Stok Barang (Inventory)

Saldo stok saat ini per produk dan gudang.

```
-- Q-ERP-09: Stok Barang
SELECT
p.sku AS "SKU",
p.name AS "Produk",
pc.name AS "Kategori",
u.name AS "Satuan Dasar",
ist.quantity_on_hand AS "Stok Tersedia",
p.buy_price AS "Harga Beli",
p.sell_price AS "Harga Jual",
(ist.quantity_on_hand * p.buy_price) AS "Nilai Stok"
FROM inventory_stock ist
JOIN products p ON p.id = ist.product_id
LEFT JOIN product_categories pc ON pc.id = p.category_id
LEFT JOIN units u ON u.id = p.base_unit_id
ORDER BY p.name;
```

Q-ERP-10 — Laporan AR Aging (Umur Piutang)

Klasifikasi piutang berdasarkan umur keterlambatan.

```
-- Q-ERP-10: AR Aging Report
SELECT
c.name AS "Pelanggan",
inv.invoice_number AS "No Faktur",
inv.date AS "Tanggal",
inv.due_date AS "Jatuh Tempo",
(inv.total - inv.amount_paid) AS "Sisa Tagihan",
(CURRENT_DATE - inv.due_date) AS "Hari Terlambat",
CASE
WHEN CURRENT_DATE <= inv.due_date THEN 'Belum Jatuh Tempo'
WHEN (CURRENT_DATE - inv.due_date) <= 30 THEN '1-30 Hari'
WHEN (CURRENT_DATE - inv.due_date) <= 60 THEN '31-60 Hari'
WHEN (CURRENT_DATE - inv.due_date) <= 90 THEN '61-90 Hari'
ELSE '> 90 Hari'
END AS "Bucket"
FROM invoices inv
JOIN customers c ON c.id = inv.customer_id
WHERE inv.type = 'sales'
AND inv.status IN ('posted', 'partial')
ORDER BY c.name, inv.due_date;
```

Q-ERP-11 — Laporan AP Aging (Umur Hutang)

Klasifikasi hutang ke supplier berdasarkan umur.

```
-- Q-ERP-11: AP Aging Report
SELECT
s.name AS "Supplier",
inv.invoice_number AS "No Faktur",
inv.date AS "Tanggal",
inv.due_date AS "Jatuh Tempo",
(inv.total - inv.amount_paid) AS "Sisa Hutang",
(CURRENT_DATE - inv.due_date) AS "Hari Terlambat",
CASE
WHEN CURRENT_DATE <= inv.due_date THEN 'Belum Jatuh Tempo'
WHEN (CURRENT_DATE - inv.due_date) <= 30 THEN '1-30 Hari'
WHEN (CURRENT_DATE - inv.due_date) <= 60 THEN '31-60 Hari'
WHEN (CURRENT_DATE - inv.due_date) <= 90 THEN '61-90 Hari'
ELSE '> 90 Hari'
END AS "Bucket"
FROM invoices inv
JOIN suppliers s ON s.id = inv.supplier_id
WHERE inv.type = 'purchase'
AND inv.status IN ('posted', 'partial')
ORDER BY s.name, inv.due_date;
```

Q-ERP-12 — Laporan Aset Tetap

Daftar aset tetap dengan nilai buku dan penyusutan.

```
-- Q-ERP-12: Aset Tetap
SELECT
a.code AS "Kode Aset",
a.name AS "Nama Aset",
ac.name AS "Kategori",
a.acquisition_date AS "Tgl Perolehan",
a.acquisition_cost AS "Harga Perolehan",
a.salvage_value AS "Nilai Residu",
a.useful_life_months AS "Masa Manfaat (Bln)",
a.depreciation_method AS "Metode",
a.accumulated_depreciation AS "Akum. Penyusutan",
(a.acquisition_cost - a.accumulated_depreciation) AS "Nilai Buku",
a.status
FROM assets a
LEFT JOIN asset_categories ac ON ac.id = a.category_id
WHERE a.is_active = true
ORDER BY ac.name, a.code;
```

Q-ERP-13 — Laporan Sales Return (Return Penjualan)

Semua return penjualan dengan produk dan nilainya.

```
-- Q-ERP-13: Sales Returns
SELECT
sr.return_number AS "No Return",
sr.date AS "Tanggal",
c.name AS "Pelanggan",
so.so_number AS "Ref SO",
p.name AS "Produk",
u.name AS "Satuan",
sri.quantity AS "Qty Return",
sri.unit_price AS "Harga",
sri.total AS "Nilai Return",
sr.status
FROM sales_returns sr
JOIN sales_return_items sri ON sri.sales_return_id = sr.id
JOIN customers c ON c.id = sr.customer_id
JOIN products p ON p.id = sri.product_id
JOIN units u ON u.id = sri.unit_id
LEFT JOIN sales_orders so ON so.id = sr.sales_order_id
ORDER BY sr.date DESC;
```

Q-ERP-14 — Neraca Saldo (Trial Balance)

Saldo debit dan kredit semua akun COA.

```
-- Q-ERP-14: Neraca Saldo
SELECT
coa.code AS "Kode Akun",
coa.name AS "Nama Akun",
coa.account_type AS "Tipe",
SUM(ji.debit) AS "Total Debit",
SUM(ji.credit) AS "Total Kredit",
(SUM(ji.debit) - SUM(ji.credit)) AS "Saldo"
FROM coa
LEFT JOIN journal_items ji ON ji.coa_id = coa.id
LEFT JOIN journals j ON j.id = ji.journal_id AND j.is_posted = true
GROUP BY coa.id, coa.code, coa.name, coa.account_type
HAVING (SUM(ji.debit) != 0 OR SUM(ji.credit) != 0)
ORDER BY coa.code;
```

Q-ERP-15 — Laporan Rekap Penjualan per Pelanggan

Total penjualan dikelompokkan per pelanggan dalam periode tertentu.

```
-- Q-ERP-15: Rekap Penjualan per Pelanggan
-- Ganti '[START]' dan '[END]' dengan tanggal, mis. '2026-01-01' dan '2026-05-31'
SELECT
c.name AS "Pelanggan",
COUNT(inv.id) AS "Jumlah Faktur",
SUM(inv.subtotal) AS "Total Subtotal",
SUM(inv.tax_amount) AS "Total PPN",
SUM(inv.total) AS "Total Penjualan",
SUM(inv.amount_paid)AS "Total Dibayar",
SUM(inv.total - inv.amount_paid) AS "Total Sisa"
FROM invoices inv
JOIN customers c ON c.id = inv.customer_id
WHERE inv.type = 'sales'
AND inv.status NOT IN ('draft','cancelled')
AND inv.date BETWEEN '[START]' AND '[END]'
GROUP BY c.id, c.name
ORDER BY SUM(inv.total) DESC;
```

3.3 Supabase JavaScript Client Queries

Alternatif SQL, gunakan Supabase JS client langsung dari React/Node.js (sudah tersedia di erp-acc via *src/lib/supabase.js*):

Setup: Install xlsx di ERP-ACC

```
# Jalankan di: apps/erp-acc/erp-app/
npm install xlsx
# Kemudian import di file yang membutuhkan:
import * as XLSX from 'xlsx'
```

Fetch Sales Orders + Detail untuk Excel

```
import { supabase } from '../lib/supabase'
import * as XLSX from 'xlsx'
export async function exportSalesOrdersToExcel() {
const { data, error } = await supabase
.from('sales_orders')
.select(`
so_number, date, status, subtotal, tax_amount, total, notes,
customer:customers(name),
items:sales_order_items(
quantity, unit_price, total,
product:products(name, sku),
unit:units(name)
)
`)
.order('date', { ascending: false })
if (error) throw error
// Flatten untuk Excel (1 baris per item)
const rows = []
data.forEach(so => {
so.items.forEach(item => {
rows.push({
'No SO': so.so_number,
'Tanggal': so.date,
'Pelanggan': so.customer?.name,
'Status': so.status,
'SKU': item.product?.sku,
'Produk': item.product?.name,
'Satuan': item.unit?.name,
'Qty': item.quantity,
'Harga': item.unit_price,
'Subtotal': item.total,
'Total SO': so.total,
})
})
})
const ws = XLSX.utils.json_to_sheet(rows)
const wb = XLSX.utils.book_new()
XLSX.utils.book_append_sheet(wb, ws, 'Sales Orders')
XLSX.writeFile(wb, 'SalesOrders.xlsx')
}
```

Fetch Invoices (Piutang) untuk Excel

```
export async function exportSalesInvoicesToExcel() {
const { data, error } = await supabase
.from('invoices')
.select(`
invoice_number, date, due_date, subtotal,
tax_amount, total, amount_paid, status,
customer:customers(name)`,)
.eq('type', 'sales')
.order('date', { ascending: false })
if (error) throw error
const rows = data.map(inv => ({
'No Faktur': inv.invoice_number,
'Tanggal': inv.date,
'Jatuh Tempo': inv.due_date,
'Pelanggan': inv.customer?.name,
'Total': inv.total,
'Sudah Dibayar': inv.amount_paid,
'Sisa': inv.total - inv.amount_paid,
'Status': inv.status,
}))
const ws = XLSX.utils.json_to_sheet(rows)
const wb = XLSX.utils.book_new()
XLSX.utils.book_append_sheet(wb, ws, 'Faktur Penjualan')
XLSX.writeFile(wb, 'FakturPenjualan.xlsx')
}
```

Export Multi-Sheet: SO + Invoices + Payments

```
export async function exportAllErpToExcel() {
const wb = XLSX.utils.book_new()
// Fetch semua data secara paralel
const [soRes, invRes, payRes] = await Promise.all([
supabase.from('sales_orders')
.select('so_number,date,status,total,customer:customers(name)')
.order('date', { ascending: false }),
supabase.from('invoices')
.select('invoice_number,date,due_date,total,amount_paid,status,customer:customers(name)')
.eq('type','sales').order('date', { ascending: false }),
supabase.from('payments')
.select('date,type,amount,notes,account:accounts(name),invoice:invoices(invoice_number)')
.order('date', { ascending: false }),
])
// Sheet 1: Sales Orders
const soRows = soRes.data.map(r => ({
'No SO': r.so_number, 'Tanggal': r.date,
'Pelanggan': r.customer?.name, 'Status': r.status, 'Total': r.total
}))
XLSX.utils.book_append_sheet(wb, XLSX.utils.json_to_sheet(soRows), 'SO')
// Sheet 2: Invoices
const invRows = invRes.data.map(r => ({
'No Faktur': r.invoice_number, 'Tanggal': r.date,
'Pelanggan': r.customer?.name, 'Total': r.total,
'Dibayar': r.amount_paid, 'Sisa': r.total - r.amount_paid, 'Status': r.status
}))
XLSX.utils.book_append_sheet(wb, XLSX.utils.json_to_sheet(invRows), 'Faktur')
// Sheet 3: Payments
const payRows = payRes.data.map(r => ({
'Tanggal': r.date, 'Tipe': r.type, 'Rekening': r.account?.name,
'Ref Faktur': r.invoice?.invoice_number, 'Jumlah': r.amount
}))
XLSX.utils.book_append_sheet(wb, XLSX.utils.json_to_sheet(payRows), 'Pembayaran')
XLSX.writeFile(wb, `ERP_Export_${new Date().toISOString().slice(0,10)}.xlsx`)
}
```

4. Template Script Export All-in-One

4.1 Node.js Script — Export Semua Data ERP-ACC ke Excel

Script ini dapat dijalankan dari command line untuk export seluruh data ERP-ACC ke satu file Excel multi-sheet. Berguna untuk backup data, laporan bulanan, atau audit trail.

```
// File: scripts/exportErpToExcel.mjs
// Jalankan: node scripts/exportErpToExcel.mjs
import { createClient } from '@supabase/supabase-js'
import * as XLSX from 'xlsx'
import { writeFileSync } from 'fs'
const supabase = createClient(
process.env.VITE_SUPABASE_URL,
process.env.SUPABASE_SERVICE_ROLE_KEY // gunakan service role untuk full access
)
async function fetchAll(table, select = '*') {
const { data, error } = await supabase.from(table).select(select)
if (error) { console.error(`Error ${table}:`, error.message); return [] }
return data
}
async function main() {
console.log('Mengambil data...')
const wb = XLSX.utils.book_new()
// Sales Orders
const so = await fetchAll('sales_orders',
'so_number,date,status,total,customer:customers(name)')
XLSX.utils.book_append_sheet(wb,
XLSX.utils.json_to_sheet(so.map(r => ({
'No SO': r.so_number, 'Tanggal': r.date,
'Pelanggan': r.customer?.name, 'Status': r.status, 'Total': r.total
}))), 'Sales Orders')
// Purchase Orders
const po = await fetchAll('purchase_orders',
'po_number,date,status,total,supplier:suppliers(name)')
XLSX.utils.book_append_sheet(wb,
XLSX.utils.json_to_sheet(po.map(r => ({
'No PO': r.po_number, 'Tanggal': r.date,
'Supplier': r.supplier?.name, 'Status': r.status, 'Total': r.total
}))), 'Purchase Orders')
// Invoices (Sales)
const sinv = await fetchAll('invoices',
'invoice_number,date,due_date,total,amount_paid,status,customer:customers(name)')
XLSX.utils.book_append_sheet(wb,
XLSX.utils.json_to_sheet(
sinv.filter(r => r.type === 'sales').map(r => ({
'No Faktur': r.invoice_number, 'Pelanggan': r.customer?.name,
'Total': r.total, 'Dibayar': r.amount_paid, 'Status': r.status
```

```
})), 'Faktur Penjualan')
// Payments
const pay = await fetchAll('payments',
'date,type,amount,notes,account:accounts(name)')
XLSX.utils.book_append_sheet(wb,
XLSX.utils.json_to_sheet(pay.map(r => ({
'Tanggal': r.date, 'Tipe': r.type,
'Rekening': r.account?.name, 'Jumlah': r.amount
})), 'Pembayaran')
// Stok
const stk = await fetchAll('inventory_stock',
'quantity_on_hand,product:products(name,sku,buy_price)')
XLSX.utils.book_append_sheet(wb,
XLSX.utils.json_to_sheet(stk.map(r => ({
'SKU': r.product?.sku, 'Produk': r.product?.name,
'Stok': r.quantity_on_hand,
'Nilai': r.quantity_on_hand * (r.product?.buy_price || 0)
})), 'Stok')
const filename = `ERP_Export_${new Date().toISOString().slice(0,10)}.xlsx`
XLSX.writeFile(wb, filename)
console.log(`Export selesai: ${filename}`)
}
main().catch(console.error)
```

4.2 Cara Menjalankan Script

```
# 1. Masuk ke direktori erp-acc
cd apps/erp-acc/erp-app
# 2. Install dependensi (jika belum)
npm install xlsx @supabase/supabase-js
# 3. Set environment variables
# Windows PowerShell:
$env:VITE_SUPABASE_URL = 'https://xxxx.supabase.co'
$env:SUPABASE_SERVICE_ROLE_KEY = 'eyJhb...'
# 4. Jalankan script
node scripts/exportErpToExcel.mjs
# Output: ERP_Export_2026-05-28.xlsx (di folder saat ini)
```

PENTING — Service Role Key: Script ini menggunakan **SUPABASE_SERVICE_ROLE_KEY** (bukan anon key) agar bisa membaca semua data tanpa batasan RLS. Jangan pernah expose key ini ke browser! Gunakan hanya di server/script lokal yang aman.

5. Catatan Keamanan & Best Practice

1. Jangan Expose API Key di Browser

Service Role Key Supabase harus hanya digunakan di server atau script lokal. Di React app (browser), gunakan hanya anon key + RLS policy yang tepat.

2. Gunakan RLS untuk Akses Data

Pastikan Row Level Security (RLS) aktif di Supabase. Export dari browser hanya akan mendapatkan data yang diizinkan oleh policy role pengguna yang sedang login.

3. Filter Data Sebelum Export

Jangan fetch semua data sekaligus jika data sudah banyak (>10.000 baris). Selalu tambahkan filter tanggal atau parameter lain untuk membatasi ukuran export.

4. Validasi Hak Akses Pengguna

Tambahkan pengecekan role sebelum memperbolehkan export. Misalnya, hanya role admin_keuangan atau owner yang dapat mengeksport data keuangan.

5. Audit Trail Export

Catat setiap aksi export di audit log — siapa yang mengeksport, kapan, dan data apa. Ini penting untuk compliance dan keamanan.

6. Enkripsi File Excel Sensitif

Untuk data sangat sensitif (data keuangan, daftar pelanggan), pertimbangkan untuk menambahkan password protection pada file Excel menggunakan library xlsx.

7. Integrasi Power Query Excel (ERP-ACC)

Supabase mendukung koneksi langsung dari Excel via Power Query menggunakan REST API. URL: https://[project-id].supabase.co/rest/v1/[table]. Tambahkan header Authorization: Bearer [anon-key].

8. Backup Berkala via Script

Jadwalkan script Node.js exportErpToExcel.mjs untuk berjalan otomatis setiap minggu atau bulan menggunakan Windows Task Scheduler atau cron job.

Ringkasan Akhir

App	Export Excel	Metode Terbaik	Effort
BUL-Accounting	Sudah ada (sebagian)	Tambah fungsi export di exportUtils.js	Rendah
ERP-ACC	Belum ada	Install xlsx + buat exportUtils.js baru	Sedang
ERP-ACC (DB)	SQL Query	Supabase SQL Editor → Export CSV → Buka di Excel	Sangat Rendah

App	Export Excel	Metode Terbaik	Effort
ERP-ACC (Live)	Power Query	Excel → Get Data → Web → Supabase REST API	Sedang

Langkah Pertama yang Direkomendasikan:
1. Untuk BUL-Accounting: tambahkan tombol Export di PenjualanPage, KasBankPage, BiayaPage.
2. Untuk ERP-ACC: jalankan Q-ERP-01 s/d Q-ERP-15 di Supabase SQL Editor untuk memverifikasi data, lalu install xlsx dan buat fungsi export.
3. Jangka panjang: hubungkan Excel ke Supabase via Power Query untuk laporan yang selalu up-to-date.